A mobile application for augmenting technical documentation using AR and AI

CS310: Third Year Project

Project Management Report

Ani Bitri

2287990

Supervisors: Dr. Paris Giampouras, John McNamara

Department of Computer Science

University of Warwick and IBM

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1 Methodology

Given the complexity and high level of development required for the project, an agile software development methodology was adopted. This drew inspiration from the Scrum framework while also incorporating elements of Kanban for task management and workflow visualization, adapted for a solo student project under academic and industrial supervision.

The project was structured into a series of fortnightly sprints, each consisting of supervisor meetings, planning, development and testing phases. The sprint planning phases involved setting clear objectives for each sprint, breaking down the work into manageable tasks and prioritizing them based on their importance. The development phase focused on implementing the planned features and functionalities, while the testing phase involved rigorous testing of the implemented features to verify their correctness and identify any issues or bugs.

Feedback from supervisors was actively sought and incorporated into the development process, ensuring that the project stayed on track and aligned with the overall objectives. Regular meetings with supervisors provided an opportunity to discuss progress, address any challenges and receive guidance on the next steps.

Trello boards were used as part of the agile approach to manage tasks, track progress and visualize the workflow. Tasks were categorized into different columns, such as "To Do", "Doing", "Testing" and "Done", allowing for a clear overview of progress. Git was used for version control, enabling efficient code management and easy rollback if necessary. Clear documentation of the codebase and development process was maintained throughout the project by writing Markdown reports and comments in the code, ensuring that the project is well documented and can be easily understood by others.

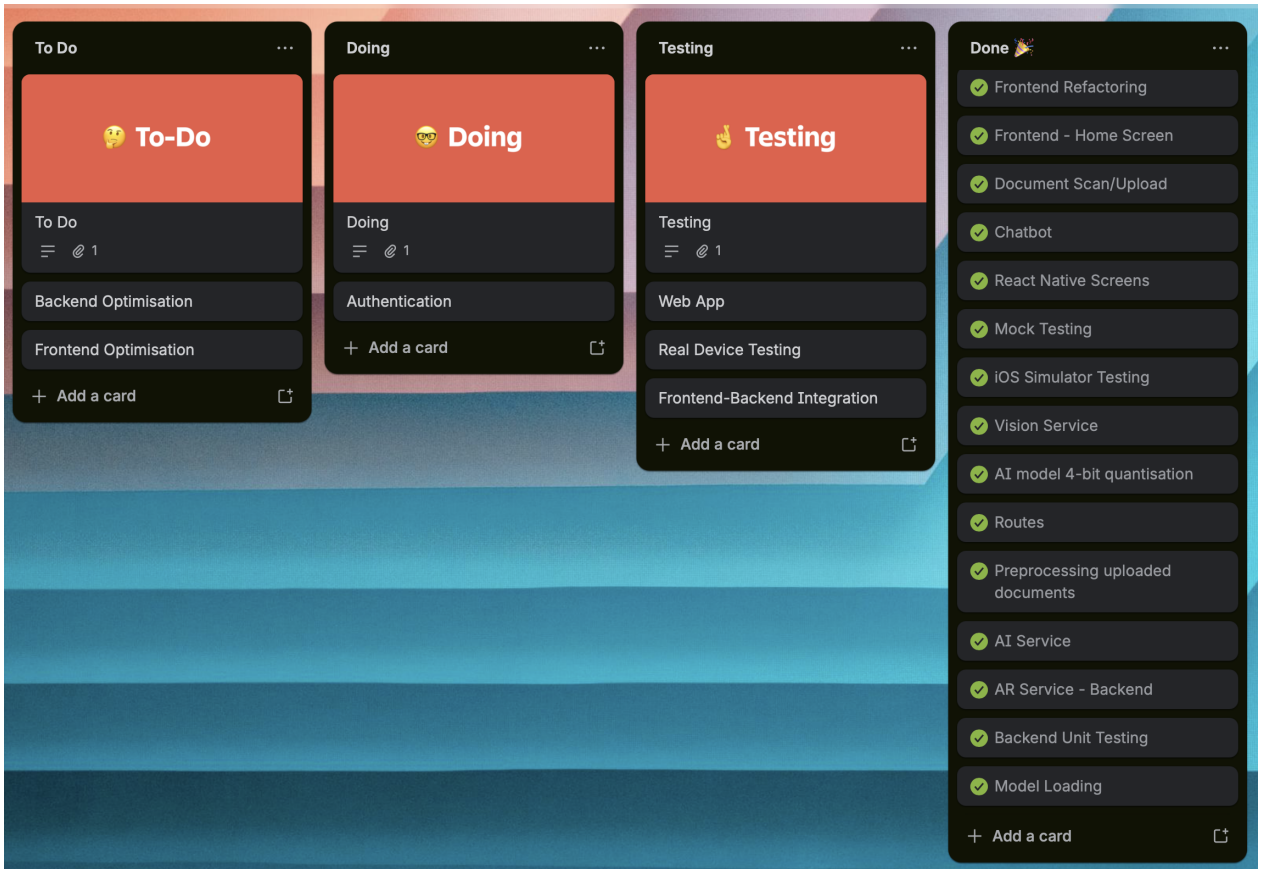

Figure 2: Trello board used for task management throughout the project.

Figure 3: Git commit history showing development activity over the project.

2 Time Management

The project schedule was planned in advance, with specific milestones and deadlines set for each phase. However, the schedule was flexible enough to accommodate changes and unforeseen challenges that arose during development. The AR phase (component detection and AR experience) took longer than expected due to the complexity of the task and the need for multiple iterations to achieve the desired results. The integration phase also took more time than anticipated, as it required careful coordination between dependencies and ensuring that all components were compatible with each other. Despite these challenges, the project was completed within the overall timeline, with an MVP delivered by the end of the project and additional features implemented in the final stages.

The unplanned web frontend was added during the frontend phase of the project, as it was deemed necessary to provide a more accessible frontend during development and testing. This addition did not significantly impact the overall timeline, as it was developed in parallel with the mobile frontend and was completed within the same timeframe. During the backend operations phase, the focus was on optimising the code and improving the performance of the backend services. After the start of the testing phase, it became clear that more time was needed for backend optimisation. As a result, the backend operations phase was extended and ran in parallel with the testing phase, allowing for continuous improvements to the backend while testing was ongoing.

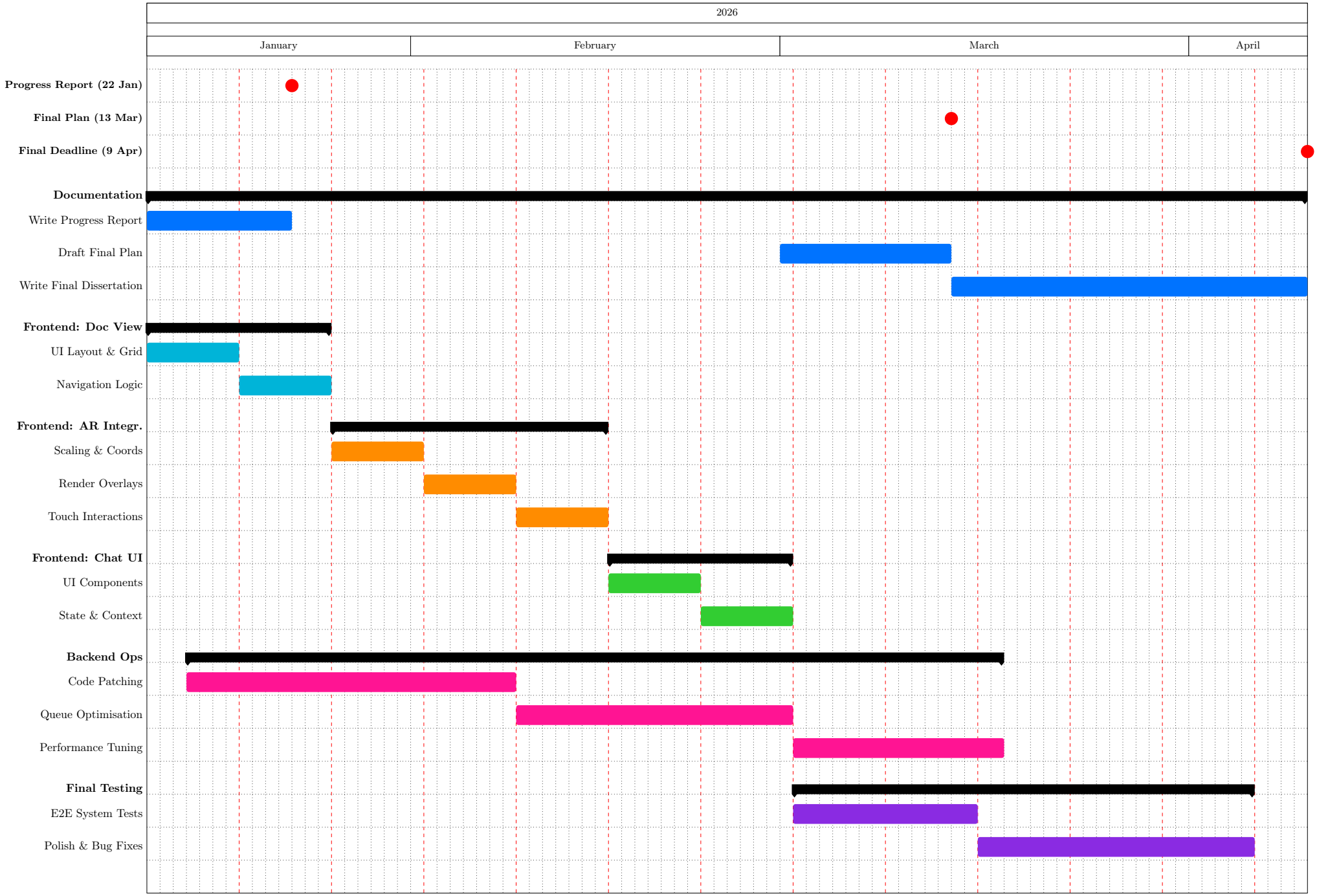

Figure 4: Revised Project Timeline

3 Risk Management

Risk	Likelihood	Impact	Mitigation	Outcome
Hardware Unavailability	Low	High	Use mock mode for testing	Did not materialise as powerful machines were available from the Department
ViroReact integration issues	Medium	High	Prototype AR rendering early, have a fallback on 2D overlays	Partially materialised, as it was not completely compatible with initial versions of the frontend, but this was resolved in later iterations by building the frontend around ViroReact after the issue was caught early
AI Model underperformance	Medium	High	Upgrade model, evaluate Granite on representative samples early and tune prompts	Not materialised as evaluation was done early, leading to tuned prompts since the beginning of the development process
Data Loss	Low	High	Frequent Git commits, model weights re-downloadable from Hugging Face, codebase present in multiple machines	Not materialised as a Git repository was maintained throughout development and the codebase was present in at least two machines at all times
Unfamiliarity with new technologies	Medium	Medium	Define a technology stack early, use known technologies, familiarise with new ones if necessary	Not materialised as it was predicted and addressed early
Scope Creep	Medium	Medium	Define clear milestones and objectives, regular supervisor meetings to stay on track	Partially materialised as the web frontend was added during development, but it was developed in parallel with the mobile frontend and did not impact the overall timeline

4 Unforeseen Challenges

An unforeseen challenge that arose during development was the incompatibility of devices and emulators between the frontend and backend components. The frontend was developed on a macOS machine and tested on iOS simulators and a physical iPhone device, while the backend was developed and tested on a Linux machine that had sufficient computational power to run the Granite model but was unsuitable for mobile development. This led to challenges in testing the end-to-end functionality of the application, as the frontend and backend components could not be tested together easily on a single machine. To address this issue, the backend server was exposed through an ngrok tunnel, allowing the frontend to communicate with backend services during development and testing, and enabling end-to-end testing of the application despite device incompatibility.

5 Professional, Ethical, Legal and Social Issues

During development, it was found that the project raised several professional and legal issues that needed to be addressed. One of the main concerns was related to data privacy and security, as the application handles potentially sensitive documents and user interactions. To address this, the backend implemented security measures and permission checks to ensure that it only accesses files within the designated input and output directories, preventing unauthorized access to the user's file system. Additionally, the API endpoints are protected by a token-based authentication mechanism to prevent unauthorized access to the backend services.

Intellectual property rights remain with IBM, as the project was developed under their supervision and using their resources. Open-source libraries and tools were used where possible, with proper credit in the deliverables and documentation.

Beyond these technical measures, no specific ethical or social issues were identified during the development of the project.